

How the Zeta Function Encodes Primes

Leonhard Euler was a Swiss polymath of the 18th century whose contributions spanned most areas of mathematics [1, p. 406] [2]. Euler's output was so extensive that it has often been said that many of his results are named after later rediscoverers to avoid overuse of his name [3, p. 87]. One of his most well-known areas of contribution is number theory, where his interest in infinite series and prime numbers laid the groundwork for analytic number theory and the zeta function [1, p. 409] [2]. This insight revealed unexpected structure within the prime numbers and showed for the first time that their distribution could be investigated analytically rather than purely arithmetically [4].

Euler's interest began with the known result that the harmonic series, $\sum_{n=1}^{\infty} \frac{1}{n}$, diverges [5, p. 241], sparking his curiosity as to whether the corresponding sum taken over the reciprocals of primes, $\sum_p \frac{1}{p}$, would also diverge [2]. At the time, divergence was not rigorously understood; before the theory of convergence had been developed by Niels Henrik Abel and Augustin-Louis Cauchy, mathematicians had little hesitation using non-convergent series in both theoretical and numerical investigations [6, p. 317]. In this context, Euler's work stands out, as he not only used such series but sought to understand their behaviour and extract meaningful results from them [2].

An intuitive approach would be to split the harmonic series into two separate sums, one over the prime numbers and the other over the composite numbers, finding which of the two is responsible for its divergence if not both. However, manipulations of divergent series are not generally valid [6]. To overcome this difficulty, Euler considered the more general sum $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, now known as the zeta function [7]. This notation was not used by Euler, but his arguments can be expressed more conveniently in this modern form rather than writing out the first several terms of each series explicitly. For $s > 1$, this series converges [8], allowing Euler to separate it into two infinite parts: $\zeta(s) = \left[1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{5^s} + \dots\right] + \left[\frac{1}{4^s} + \frac{1}{6^s} + \frac{1}{8^s} + \dots\right]$.

Euler's proof started by considering the reciprocal prime sum, which in modern notation can be denoted by $A = \sum_p \frac{1}{p}$, together with similar sums with higher powers such as $B = \sum_p \frac{1}{p^2}$, $C = \sum_p \frac{1}{p^3}$ and so on [9]. Using logarithmic expansions and infinite series, Euler showed that the expression $A + \frac{1}{2}B + \frac{1}{3}C + \frac{1}{4}D + \dots$ can be written as the logarithm of an infinite product over all prime numbers, whose behaviour corresponds to the harmonic series [9]. In modern notation, this relationship can be expressed as $\log\left(\prod_p \frac{1}{1 - \frac{1}{p}}\right) = \sum_p \frac{1}{p} + (\text{finite terms})$ [7]. Euler further argued that the product corresponds to the harmonic series, which diverges [9]. Since the additional terms involving B, C, D, ... are finite, the divergence comes from the sum over primes. Thus, $\sum_p \frac{1}{p}$ also diverges, a striking result demonstrating that although primes become increasingly sparse, they remain prevalent enough to produce a divergent sum and providing another proof of Euclid's theorem of infinite prime numbers [8].

In this proof, Euler obtained what is now known as the Euler product, revealing a fundamental connection between sums over integers and products over primes [2, p. 67]. In his original reasoning, he began with the harmonic series $x = 1 + \frac{1}{2} + \frac{1}{3} + \dots$, and, although x is infinite, began informally filtering terms divisible by successive prime numbers [9]. First, he wrote $x - \frac{1}{2}x = 1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{9} + \dots$, which removes all even denominators [9]. Continuing this idea, Euler found $\frac{1}{2}x - \frac{1}{3}\left[\frac{1}{2}x\right] = 1 + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \frac{1}{13} + \dots$, which Euler interpreted as corresponding to $\frac{1 \cdot 2}{2 \cdot 3}x$, reflecting the removal of multiples

of 3 [9]. Extending this infinitely, Euler concluded that this would informally lead to

$\frac{1 \cdot 2 \cdot 4 \cdot 6 \cdot 10 \cdot 12 \cdot 16 \cdot \dots}{2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot \dots} x = 1$, linking the harmonic series and an infinite product over primes involving

factors of the form $\frac{p}{p-1} = \frac{1}{1-\frac{1}{p}}$, leading to the Euler product $\prod_p \frac{1}{1-\frac{1}{p}} = \sum_{n=1}^{\infty} \frac{1}{n}$ [2, p. 69]. For a modern

reader, repeated operations on divergent series lack validity [6]. However, the argument had historical significance and led Euler to formulate the Euler product representation associated with the divergence of the harmonic series, which can be proven using modern methods by using the fundamental theorem of arithmetic to express every number in the form $2^a 3^b 5^c 7^d \dots$ [10, p. 3] and

splitting the harmonic series like so $x = \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots\right] * \left[1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots\right] *$

$\left[1 + \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \dots\right] * \dots$ [2, p. 69] then using the geometric series formula to get $x = \frac{1}{1-\frac{1}{2}} * \frac{1}{1-\frac{1}{3}} *$

$\frac{1}{1-\frac{1}{5}} * \dots$ as seen in the Euler product formula's modern notation [2, p. 69]. This was later made rigorous

with the development of analytic number theory, and in 1876 Leopold Kronecker proved that

$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$ for $s > 1$, interpreting Euler's theorem as the limiting case as $s \rightarrow 1^+$ [11, p. 413],

so putting aside Euler's rigor, we can see that his powerful intuition had made a connection between the harmonic series and prime numbers and the consequences of this discovery were vast [2].

The zeta function was transformed through the work of Bernhard Riemann, who extended the definition to a complex variable in 1859 [12]. He proved it has a meromorphic continuation, meaning it extends to a meromorphic function on the complex plane with a simple pole at $s=1$ [7, p. 15]. Riemann also established the functional equation relating $\zeta(s)$ to $\zeta(1-s)$, revealing a symmetry around the line $Re(s) = \frac{1}{2}$ and connecting the behaviour of the function on the right half-plane to the left [7, p. 13].

Using complex analysis, he related the zeros of the zeta function to the distribution of prime numbers using properties of the logarithmic derivative, which reflects the zeros and poles of a function [13] [14, p. 312]. Since there is only a simple pole at $s=1$, the non-trivial zeros are essential in determining whether the primes follow a well-behaved curve with small oscillations or if there are larger irregularities in how primes are spaced [7]. The most famous unsolved problem in mathematics arose from these zeros, this being the Riemann hypothesis, which predicts that all non-trivial zeros of the Riemann zeta function lie on the line with real part $\frac{1}{2}$ [15].

The ideas of Euler and Riemann culminated in the Prime Number Theorem, first conjectured independently by Carl Friedrich Gauss and Adrien-Marie Legendre [1, p. 448 and 467], which describes the asymptotic distribution of prime numbers. In modern notation, it states that the number of primes less than x satisfies $\pi(x) \sim \frac{x}{\log x}$, as $x \rightarrow \infty$, showing that primes become less frequent in a predictable way [16, p. 449]. Proofs of this result relied heavily on analytic properties of the zeta function, particularly its behaviour in the complex plane and the absence of zeros on the line $Re(s) = 1$ [8, p. 7]. Euler's product formula provided the fundamental link between primes and analytic expressions [16, p. 442], while Riemann's extension of the zeta function into the complex plane made it possible to study prime numbers using methods from complex analysis [16, p. 442].

Although the Riemann hypothesis remains unproven, it would provide even more precise information about the distribution of primes, further illustrating the depth of the first connection to the primes by Euler. From Euler's initial insight linking primes to infinite series, through Riemann's extension into the complex plane, to results such as the prime number theorem, the zeta function has become a central tool in the study of prime numbers. These developments demonstrate how a single analytic idea can reveal profound structure within the integers and continue to shape modern analytic number theory.

References

- [1] Boyer, C. B. and Merzbach, U. C. (2011) *A History of Mathematics*, 3rd ed., Hoboken: Wiley.
- [2] Dunham, W. (1999) *Euler: The Master of Us All*, Washington, DC: Mathematical Association of America.
- [3] Richeson, D. S. (2008) *Euler's Gem: The Polyhedron Formula and the Birth of Topology*, Princeton, NJ: Princeton University Press.
- [4] Stopple, J. (2003) *A Primer of Analytic Number Theory: From Pythagoras to Riemann*, Cambridge: Cambridge University Press.
- [5] Lewin, J. and Lewin, M. (1993) *An Introduction to Mathematical Analysis*, 2nd ed., New York: McGraw-Hill.
- [6] Bromwich, T. J. l'A. (1908) *An Introduction to the Theory of Infinite Series*, London: Macmillan.
- [7] Titchmarsh, E. C. (1986) *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford: Clarendon Press.
- [8] Patterson, S. J. (1988) *An Introduction to the Theory of the Riemann Zeta-Function*, Cambridge: Cambridge University Press.
- [9] Euler, L. (1744) *Variae observationes circa series infinitas*, in *Commentarii Academiae Scientiarum Petropolitanae*, 9, pp. 160–188; reprinted in *Opera Omnia*, Series I, Vol. 14, pp. 217–244. English translation by P. Viader and L. Bibiloni.
- [10] Hardy, G. H. and Wright, E. M. (1938) *An Introduction to the Theory of Numbers*. 1st ed. Oxford: Clarendon Press.
- [11] Dickson, L. E. (1919–1923) *History of the Theory of Numbers*. Washington, DC: Carnegie Institution of Washington.
- [12] Dittrich, W. (2017) *On Riemann's Paper, "On the Number of Primes Less Than a Given Magnitude"*. Institut für Theoretische Physik, Universität Tübingen, Tübingen, Germany.
- [13] Deshpande, J. V. (1987) *Complex Analysis*. New York: McGraw-Hill Education.
- [14] Priestley, H. A. (2003) *Introduction to Complex Analysis*. 2nd ed. Oxford: Oxford University Press.
- [15] Chowla, S. (1965) *The Riemann Hypothesis and Hilbert's Tenth Problem*. New York: Gordon and Breach Science Publishers.
- [16] Lang, S. (1999) *Complex Analysis*. 4th ed. New York: Springer.